Supplement for VERA: Support-Aware Certification of Representation Edits Under Deployment Shift

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Abstract

This supplement provides full proofs, protocol details, implementation audits, and receipt-level reporting for VERA. It follows the claims and terminology of the main paper. Standard finite-family testing, exact binomial intervals, CVaR duality, and bounded-density-ratio robustness are credited as prior machinery; the proposed object is the support-aware representation-edit shift envelope for paired representation interventions and a registered retrained-attacker portfolio, together with prospective evidence allocation across its limiting supported audit cells.

1 Scope and Reproducibility Contract

The exploratory parent preregistration is identified by SHA-256 `4c9343207f940220d762e230023531af` `353cc0886a982f133ff7794c87f13ab7`. It governs seeds 0–4, which were used only for estimand and power diagnostics. The untouched-seed balanced confirmatory preregistration is identified by SHA-256 `2767b64f1fa844f512026c5cc4d5e81b` `ca4ed92bef9c8dfd5287dec2918395aa` and immutable Git tag `vera-balanced-confirmatory-locked-2026-07-14`. The untouched-seed secondary-ablation protocol is identified by SHA-256 `389b32fe926bf427abc80b4b249c6e9f` `27342b1ade3ecfc19e71fee32b1362a6`. The descriptive real learning-curve diagnostic is separately fixed by SHA-256 `b2f0a5c66c5fa55c84aeeaa30f397820` `4a4217d8c50e60973a671a21d316e824`. The candidate-family/group-count coverage extension is fixed by SHA-256 `c63a4a5b68ab194dd52bbaea1f1d3266` `a450147349e9994a51cd0566fdf61b2e`. The independent disjoint-seed stress replication is fixed by SHA-256 `348c2784ec8d6bc9cef3c8449dccd4a0` `7c394f1e3b54f8231b7c9c88b52caad3`; it repeats the official-code matrix on seeds 13–44 with one locked stress contract per supported dataset. The prospective controlled-shift study is fixed by SHA-256 `ffcb4d44d556e8acab720b8df55ff291` `78e7b2d3d36d20f9bb686906f864dfee`; it uses fresh seeds 45–108, disjoint from every design and prior confirmatory seed. A prospectively frozen interpretation note, SHA-256 `7fcbdd47c80112dd5a995f60c9a04f03` `8aaf4e715b19a68f68594c65cbfb6590`, identifies parent-study keys that were retained by the JSON generator but are inactive for seeds 45–108. It changes no controlled-study setting or success condition. The earlier balanced confirmatory lock fixes datasets, methods, candidates, seeds 5–12, splits, thresholds, attackers, validation fractions, that study's primary IUT at $\Gamma = 1$, the $\Gamma = 1.01$ sensitivity, the simultaneous envelope, and every pass condition before the first new-seed run.

All reported real runs are tied to one parent repository commit and one pinned upstream eraser commit. Each compact JSON receipt names its split indices, preprocessing policy, official upstream entry point, candidate strengths, and the SHA-256 digest of every external-drive audit archive. The audit opens every archive and verifies required arrays, lengths, supports, attacker names, candidate uniqueness, repository cleanliness, and commit identity. A missing, proxy, dirty, or hash-mismatched row fails closed.

2 Closest-Work Capability Matrix

Tables 1–4 compare reported outputs and guarantees, not components that could be assembled by combining papers. "P" means partial, variant-specific, or materially different; "–" means that the capability is not reported. This conservative coding credits generic machinery to prior work and isolates the proposed scientific object. The evidence-allocation extension is deliberately not coded as another binary column: its additive fixed-budget minimax DKW-slack objective must be positioned separately against active testing, stratified and Neyman allocation, shared-observation experimental design, and generic convex resource allocation. Its square-score and two-thirds-power special cases are not claimed as generic allocation novelty.

The columns are finite selection (FS), multiple risks (MR), paired intervention harm (PH), fresh attacker retraining (FR), multiple attacker families (MA), environment-conditional robustness (EC), source-conditional robustness (SC), arbitrary environment mixtures (EM), source-prevalence invariance (SP), a continuum of budgets (CB), a vector profile (VP), a common radius (CR), support mismatch detection (SM), whole-intervention abstention (WA), post-inspection validity (PI), and no target-deployment sample (NT).

IID certification can accept an edit that VERA rejects. Let leakage be Bernoulli with population recall $p = .04$ under a .05 contract. Even without sampling uncertainty, the IID contract passes. At $\Gamma = 2$, the bounded-reweighting risk is $\min(1, \Gamma p) = .08$, so VERA rejects the declared shift. The distinction is therefore in the population contract, not merely in confidence-interval width.

Worst-group evaluation does not return the envelope. Suppose two observed environments have Bernoulli harm rates .04 and .03 under a .05 threshold. IID worst-group harm is .04, but the robust risks are $\min(1, .04\Gamma_1)$ and $\min(1, .03\Gamma_2)$. VERA returns coordinate radii 1.25 and 1.67, their common radius 1.25, and the limiting environment. A scalar worst-group value cannot answer whether a later-declared profile such as $(\Gamma_1, \Gamma_2) = (1.2, 1.5)$ is covered.

A finite attacker portfolio is not universal erasure. Let U, V be independent fair bits and encode $S = U \text{ XOR } V$ in the edited representation (U, V). Linear attackers have balanced accuracy $1/2$, while an XOR attacker recovers S perfectly. VERA simultaneously controls only the registered attacker portfolio and never converts that finite statement into absence of all sensitive information.

3 Shift-Robust Paired Certification

Let P denote the certification reference distribution and let $V : \mathcal{X} \to [a, b]$ be a bounded audit variable. In the target-preservation application, $V = H_e$ is the paired increase in loss caused by edit e relative to the identity edit on the same example. For $\Gamma \geq 1$, define the ambiguity set

$$\mathcal{Q}_\Gamma(P) = \left\{ Q \ll P : 0 \leq \frac{\mathrm{d}Q}{\mathrm{d}P} \leq \Gamma \quad P\text{-almost surely} \right\} \tag{1}$$

and the robust risk $\rho_{\Gamma,P}(V) = \sup_{Q \in \mathcal{Q}_\Gamma(P)} \mathbb{E}_Q[V]$. The reference law is contract-specific. Target contracts use the environment-conditional law P_g. Leakage uses P_s, the law conditional on represented source class $s \in \{0, 1\}$, and evaluates equal-weight robust class recall rather than ordinary accuracy. Accordingly, an admissible deployment profile requires $Q_g \in \mathcal{Q}_{\gamma_g}(P_g)$ for target harm and $Q_s \in \mathcal{Q}_{\eta_s}(P_s)$ for leakage. Environment mixture weights may vary arbitrarily. Source-class prevalences may also vary arbitrarily because the leakage estimand fixes the two class weights at one half; this does not imply control of prevalence-weighted ordinary accuracy.

Lemma 1 (Reweighting–CVaR identity)**.** *For every bounded V and $\Gamma \geq 1$,*

$$\rho_{\Gamma,P}(V) = \inf_{\eta \in [a,b]} \{\eta + \Gamma\, \mathbb{E}_P[(V - \eta)_+]\}. \tag{2}$$

Thus $\rho_{\Gamma,P}$ is upper-tail conditional value at risk at level $1 - 1/\Gamma$.

Proof. Write $w = \mathrm{d}Q/\mathrm{d}P$. The primal problem maximizes $\mathbb{E}_P[wV]$ subject to $0 \leq w \leq \Gamma$ and $\mathbb{E}_P[w] = 1$. Introducing a multiplier η for the equality constraint and maximizing pointwise over w gives $\eta + \Gamma\mathbb{E}_P[(V - \eta)_+]$. Strong duality follows because $w \equiv 1$ is feasible and the bounded linear program has a finite optimum. The optimizer places the largest permitted mass on the upper tail of V. $\square$

Given independent certification observations $V_1, \ldots, V_n$, let $\widehat{P}_n$ be their empirical distribution and let

$$\widehat{\rho}_{\Gamma,n}(V) = \inf_{\eta \in [a,b]} \left\{ \eta + \frac{\Gamma}{n} \sum_{i=1}^{n} (V_i - \eta)_+ \right\}. \tag{3}$$

Probability space and protocol contract. Let $\mathcal{F}_0$ contain everything fixed before certification: the construction data and randomness, the finite candidate edits, whether identity is selectable, target probes, registered attackers, audit variables, required environment and source cells, thresholds, confidence level, candidate utility rule, and the ambiguity model and profile caps (but not the eventual profile chosen inside a reported envelope). Conditional on $\mathcal{F}_0$, certification examples are independent draws from the declared reference law. Within a represented cell, their audit values are therefore i.i.d. from its conditional law; validity may be conditioned on the realized cell count. Candidates and components may share examples, and no independence across candidates, environments, or attackers is assumed. Probability below is over the certification draw and any protocol randomness not already included in $\mathcal{F}_0$. The guarantees hold conditionally for every admissible realization of $\mathcal{F}_0$ and hence also unconditionally by iterated expectation. A candidate, attacker, threshold, or screening rule chosen using certification outcomes is outside this contract unless its adaptivity is explicitly covered by the confidence construction.

Theorem 1 (Uniform robust paired-edit certificate)**.** *Fix K bounded audit variables before observing their certification folds. Variable k has range $[a_k, b_k]$ and n_k independent observations. Define*

$$\epsilon_k = \sqrt{\frac{\log(2K/\delta)}{2n_k}}, \qquad U_k(\Gamma) = \min\left\{b_k, \widehat{\rho}_{\Gamma,n_k}(V_k) + \Gamma(b_k - a_k)\epsilon_k\right\}. \tag{4}$$

Then, with probability at least $1 - \delta$, simultaneously for all k and all $\Gamma \geq 1$, and all $Q \in \mathcal{Q}_\Gamma(P_k)$, $\mathbb{E}_Q[V_k] \leq U_k(\Gamma)$. Consequently, any data-dependent edit selected only when its paired target-harm bound is at most τ and all of its preregistered attacker-leakage bounds are at most λ satisfies those contracts for every allowed deployment distribution.

Proof. For one variable, the Dvoretzky–Kiefer–Wolfowitz inequality gives $\sup_t |F_n(t) - F(t)| \leq \epsilon_k$ except on an event of probability δ/K. For every $\eta \in [a_k, b_k]$,

$$\begin{aligned} &\left| \mathbb{E}_P[(V - \eta)_+] - \mathbb{E}_{\widehat{P}_n}[(V - \eta)_+] \right| \\ &= \left| \int_\eta^{b_k} (F_n(t) - F(t))\,\mathrm{d}t \right| \\ &\leq (b_k - a_k)\epsilon_k. \end{aligned} \tag{5}$$

Method family	FS	MR	PH	FR
Learn Then Test (Angelopoulos et al. 2025)	Y	Y	–	–
Pareto Testing (Laufer-Goldshtein et al. 2023)	Y	Y	–	–
Conformal Risk Control (Angelopoulos et al. 2024)	P	P	–	–
Prompt Risk Control (Zollo et al. 2024)	Y	Y	–	–
Robust validation (Cauchois et al. 2024)	–	P	–	–
Robust conformal inference (Ai and Ren 2024)	P	P	–	–
CVaR risk certification (Thomas and Learned-Miller 2019)	P	P	–	–
Distributional fairness certification (Kang et al. 2022)	–	P	–	P
Wasserstein fairness auditing (Ehyaei, Farnadi, and Samadi 2025)	–	P	–	P
Selective prediction (El-Yaniv and Wiener 2010; Geifman and El-Yaniv 2019)	P	P	–	–
Group-robust evaluation (Sagawa et al. 2020; Koh et al. 2021)	–	P	–	–
Representation erasure (Ravfogel et al. 2020, 2022a; Belrose et al. 2023)	P	P	P	P
VERA	**Y**	**Y**	**Y**	**Y**

Table 1: Closest-work matrix, part I: selection and intervention structure.

Method family	MA	EC	SC	EM
Learn Then Test (Angelopoulos et al. 2025)	–	–	–	–
Pareto Testing (Laufer-Goldshtein et al. 2023)	–	–	–	–
Conformal Risk Control (Angelopoulos et al. 2024)	–	P	–	–
Prompt Risk Control (Zollo et al. 2024)	–	P	–	–
Robust validation (Cauchois et al. 2024)	–	P	–	–
Robust conformal inference (Ai and Ren 2024)	–	P	–	–
CVaR risk certification (Thomas and Learned-Miller 2019)	–	P	P	P
Distributional fairness certification (Kang et al. 2022)	P	P	P	P
Wasserstein fairness auditing (Ehyaei, Farnadi, and Samadi 2025)	P	P	P	P
Selective prediction (El-Yaniv and Wiener 2010; Geifman and El-Yaniv 2019)	–	–	–	–
Group-robust evaluation (Sagawa et al. 2020; Koh et al. 2021)	–	Y	P	Y
Representation erasure (Ravfogel et al. 2020, 2022a; Belrose et al. 2023)	P	–	P	–
VERA	**Y**	**Y**	**Y**	**Y**

Table 2: Closest-work matrix, part II: attacker and conditional-shift structure.

Applying this inequality inside Lemma 1 and taking the infimum over η yields $\rho_{\Gamma,P}(V) \leq \widehat{\rho}_{\Gamma,n}(V) + \Gamma(b_k - a_k)\epsilon_k$. A union bound over K audit variables proves simultaneous validity. On this event, every accepted edit obeys each contract; selection among accepted edits does not change the event. □

Corollary 1 (Exact discrete paired certificate)**.** *Let $L \in \{0,1\}$ have success probability p, and let $H \in \{-1,0,1\}$ have probabilities (p_-, p_0, p_+). Then*

$$\rho_{\Gamma,P}(L) = \min\{1, \Gamma p\}, \tag{6}$$

and, writing $a = \Gamma p_+$ and $d = \Gamma(1-p_-)$,

$$\rho_{\Gamma,P}(H) = \begin{cases} 1, & a \geq 1, \\ a, & a < 1 \leq d, \\ a + d - 1, & d < 1. \end{cases} \tag{7}$$

For a registered family of K contracts, replace p by a one-sided Clopper–Pearson upper bound at error δ/K. For paired harm, let U_+ be a one-sided upper bound for p_+ and L_- a one-sided lower bound for p_-, each at error $\delta/(2K)$, and substitute $\widetilde{p}_+ = \min\{U_+, 1 - L_-\}$ and $\widetilde{p}_- = L_-$ in the formula. The clipping preserves the probability-simplex constraint and can only tighten the upper bound. These substitutions give simultaneous upper confidence curves for every $\Gamma \geq 1$. Inverting the curves therefore gives the same shift-radius and false-acceptance guarantees as Theorem 4.

Proof. The worst density ratio assigns weight Γ first to value one, then to value zero, and finally assigns the residual unit mass to value minus one. This greedy allocation yields the displayed formulas. The Bernoulli expression is increasing in p. The paired expression is increasing in p_+ and decreasing in p_- in every branch. On the joint confidence event, $p_+ \leq U_+$ and $p_- \geq L_-$. Since also $p_+ \leq 1 - p_- \leq 1 - L_-$, $p_+ \leq \widetilde{p}_+$; hence the clipped substitution dominates the true robust risk. A union bound gives simultaneous validity. The coverage event is independent of Γ, so it holds over the entire continuum and remains valid after inversion and edit selection. □

For attacker a, let $L_{e,a} = \mathbf{1}\{a_e(Z^e) = S\}$ and define the balanced robust leakage

$$B_{e,a}(\eta_0, \eta_1) = \frac{1}{2} \sum_{s \in \{0,1\}} \rho_{\eta_s, P_s}(L_{e,a}). \tag{8}$$

If $U_{e,a,s}(\eta_s)$ is an upper confidence curve for the class-s recall, then $\widehat{B}_{e,a} = \frac{1}{2}(U_{e,a,0} + U_{e,a,1})$ upper-bounds Eq. (8)

Method family	SP	CB	VP	CR
Learn Then Test (Angelopoulos et al. 2025)	–	–	–	–
Pareto Testing (Laufer-Goldshtein et al. 2023)	–	–	–	–
Conformal Risk Control (Angelopoulos et al. 2024)	–	–	–	–
Prompt Risk Control (Zollo et al. 2024)	–	–	–	–
Robust validation (Cauchois et al. 2024)	–	P	–	–
Robust conformal inference (Ai and Ren 2024)	–	P	–	–
CVaR risk certification (Thomas and Learned-Miller 2019)	–	P	–	P
Distributional fairness certification (Kang et al. 2022)	P	P	–	–
Wasserstein fairness auditing (Ehyaei, Farnadi, and Samadi 2025)	P	P	–	–
Selective prediction (El-Yaniv and Wiener 2010; Geifman and El-Yaniv 2019)	–	–	–	–
Group-robust evaluation (Sagawa et al. 2020; Koh et al. 2021)	P	–	–	–
Representation erasure (Ravfogel et al. 2020, 2022a; Belrose et al. 2023)	P	–	–	–
VERA	**Y**	**Y**	**Y**	**Y**

Table 3: Closest-work matrix, part III: prevalence, budget continuum, and envelope geometry.

Method family	SM	WA	PI	NT
Learn Then Test (Angelopoulos et al. 2025)	–	P	Y	Y
Pareto Testing (Laufer-Goldshtein et al. 2023)	–	P	Y	Y
Conformal Risk Control (Angelopoulos et al. 2024)	–	–	P	P
Prompt Risk Control (Zollo et al. 2024)	P	P	Y	P
Robust validation (Cauchois et al. 2024)	–	–	P	Y
Robust conformal inference (Ai and Ren 2024)	P	–	P	P
CVaR risk certification (Thomas and Learned-Miller 2019)	–	P	P	Y
Distributional fairness certification (Kang et al. 2022)	P	–	P	Y
Wasserstein fairness auditing (Ehyaei, Farnadi, and Samadi 2025)	P	–	P	Y
Selective prediction (El-Yaniv and Wiener 2010; Geifman and El-Yaniv 2019)	–	–	P	Y
Group-robust evaluation (Sagawa et al. 2020; Koh et al. 2021)	P	–	–	Y
Representation erasure (Ravfogel et al. 2020, 2022a; Belrose et al. 2023)	–	P	–	Y
VERA	**Y**	**Y**	**Y**	**Y**

Table 4: Closest-work matrix, part IV. WA excludes methods that reject only individual examples. NT refers to the shift guarantee; methods requiring an unlabeled target sample receive P rather than Y.

on the intersection of the two coverage events. A component-level error budget α is therefore obtained by using one-sided Clopper–Pearson bounds at $\alpha/2$ for the two recalls.

Lemma 2 (Source-prior invariance)**.** *Let the deployment model declare both source-conditional laws and let source prevalence be any $\pi \in [0,1]$, including 0 or 1. Suppose the two source-conditional deployment laws obey $Q_s \in \mathcal{Q}_{\eta_s}(P_s)$. The balanced robust leakage in Eq.* (8) *is independent of π and is at most λ whenever $\widehat{B}_{e,a} \le \lambda$. A constant binary attacker has balanced leakage $1/2$ at $\eta_0 = \eta_1 = 1$, regardless of which class it always predicts.*

Proof. The estimand assigns weight one half to each conditional recall by definition, so prevalence π does not enter it. On the joint confidence event, each robust recall is at most its corresponding upper curve; averaging gives the first claim. A constant attacker has class recalls $(1,0)$ or $(0,1)$ under IID evaluation, whose equal-weight mean is $1/2$. At $\pi \in \{0,1\}$, the unused conditional remains part of the declared standardization model; the balanced functional is not ordinary accuracy observable from a deployment sample containing only one class. □

Theorem 2 (Fixed-profile candidate intersection–union test)**.** *Fix M candidate edits before certification. For candidate e, let $\{R_{e,j} \le c_j\}_{j \in \mathcal{J}_e}$ contain every target and balanced leakage component at one declared shift profile. Suppose each reported component bound $U_{e,j}$ undercovers $R_{e,j}$ with probability at most $\alpha = \delta/M$. For paired target harm, the positive- and negative-count bounds split this α equally; for balanced leakage, its two class-recall bounds do the same. Accept candidate e only if $U_{e,j} \le c_j$ for every j, then select any accepted candidate by a prespecified measurable utility rule, which may use certification values. The probability of deploying any candidate that violates at least one component contract is at most δ. If identity is itself a selectable deployment action, it is included among the M candidates; using identity only to define paired harm does not add a selection hypothesis.*

Proof. For each unsafe candidate e, choose one population-violated component $j(e)$ before the certification draw. False acceptance of e implies $U_{e,j(e)} \le c_{j(e)} < R_{e,j(e)}$, an undercoverage event of probability at most α. No union bound across components is needed because acceptance requires every component to pass: this is an intersection–union test. A union bound over the at most M unsafe candidates gives

$M\alpha = \delta$. Selection among accepted candidates is contained in the same event. $\square$

Corollary 2 (Worst-group mixture shift). *Let P_g be certified group-conditional reference distributions for $g \in \mathcal{G}$. If $\rho_{\Gamma,P_g}(V_e) \leq c$ is simultaneously certified for every edit e and group g, then for every collection $Q_g \in \mathcal{Q}_\Gamma(P_g)$ and every deployment mixture $Q_\pi = \sum_g \pi_g Q_g$ with π in the probability simplex, $\mathbb{E}_{Q_\pi}[V_e] \leq c$.*

Proof. $\mathbb{E}_{Q_\pi}[V_e] = \sum_g \pi_g \mathbb{E}_{Q_g}[V_e] \leq \sum_g \pi_g c = c$. $\square$

Let $U^T_{e,g}(\gamma_g)$ be a simultaneous upper curve for paired target harm in observed environment g, and let $U^L_{e,a,s}(\eta_s)$ be one for attacker a's class-s recall. The *support-aware representation-edit shift envelope* is

$$\widehat{\mathcal{E}}_e = \{(\boldsymbol{\gamma}, \boldsymbol{\eta}) : U^T_{e,g}(\gamma_g) \leq \tau \quad \forall g, \\ \tfrac{1}{2} \textstyle\sum_s U^L_{e,a,s}(\eta_s) \leq \lambda \quad \forall a\}. \tag{9}$$

Profiles that place deployment mass on an unobserved required environment are excluded. A missing source class excludes balanced-leakage certification unless a structural assumption identifies its conditional recall.

Theorem 3 (Support-aware representation-edit shift envelope). *With probability at least $1-\delta$, simultaneously for every registered edit e, every deployment profile in its certified envelope satisfies every registered target and balanced-leakage contract. Environment mixture weights and source-class prevalences may be chosen arbitrarily after certification. Profiles may also be chosen after inspecting the envelope without another multiplicity correction. A required target environment or source class with zero certification observations is excluded rather than extrapolated.*

Proof. The simultaneous event used for the individual curves already ranges over every edit, target environment, source class, and attacker, and each curve is valid for every value of its density-ratio argument. On that event, membership in Eq. (9) controls target harm for every allowed Q_g and balanced leakage for every allowed Q_s. Arbitrary environment mixing preserves the target threshold by Corollary 2; source prevalence does not enter Eq. (8). Inspecting or intersecting objects valid on this same event does not spend additional error probability. An unobserved required conditional law has no confidence curve and therefore contributes no admissible envelope coordinate. $\square$

Corollary 3 (Common radius and post-certification profile choice). *Under Theorem 3, the largest certified common budget is*

$$\widehat{\Gamma}_e = \sup\{\Gamma \geq 1 : (\Gamma \mathbf{1}_{|\mathcal{G}|}, \Gamma, \Gamma) \in \widehat{\mathcal{E}}_e\}, \tag{10}$$

with value zero if the IID profile fails. No additional multiplicity correction is required to inspect the envelope or choose a contained profile after certification.

Proof. Restrict Theorem 3 to the diagonal profile. Both the diagonal supremum and any later anisotropic profile choice are functions of the same simultaneous confidence event. $\square$

Corollary 4 (False acceptance). *Under the conditions of Theorem 1, the probability that VERA accepts an edit violating at least one declared robust contract is at most δ.*

For edit e, let $\mathcal{K}_e$ index its paired target-harm and registered leakage contracts, with thresholds c_k. Define the population representation-edit shift radius

$$\Gamma^\star_e = \sup\{\Gamma \geq 1 : \rho_{\Gamma,P_k}(V_k) \leq c_k \text{ for every } k \in \mathcal{K}_e\}, \tag{11}$$

with $\Gamma^\star_e = 0$ when a contract fails at $\Gamma = 1$. Let $\widehat{\Gamma}_e$ replace each population robust risk by the upper bound in Theorem 1, and use the same zero convention.

Theorem 4 (Certified representation-edit shift radius). *With probability at least $1-\delta$, simultaneously for every registered edit e, $\widehat{\Gamma}_e \leq \Gamma^\star_e$. The statement holds uniformly over the continuum of $\Gamma \geq 1$; consequently, a user may select any deployment budget $\Gamma \leq \widehat{\Gamma}_e$ after observing the certificate without an additional multiplicity correction. If computation is capped at $\Gamma_{\max}$, equality $\widehat{\Gamma}_e = \Gamma_{\max}$ is reported as right-censored.*

Proof. The DKW event in Theorem 1 does not depend on Γ. On that event, for every k and every $\Gamma \geq 1$, the reported upper curve dominates $\rho_{\Gamma,P_k}(V_k)$. Hence every Γ admitted by all empirical upper curves is admitted by all population contracts. Taking the supremum proves $\widehat{\Gamma}_e \leq \Gamma^\star_e$ for every edit on the same simultaneous event. Robust risk is non-decreasing in Γ because the ambiguity sets are nested, so the feasible budgets form an interval and bisection is valid. Post-certificate choice within that interval leaves the event unchanged. $\square$

Theorem 5 (Sufficient certification sample size). *Fix a component with range length $R = b - a$, declared budget Γ_0, and threshold c. Construct its DKW upper curve with component undercoverage probability α. If its population robust risk obeys $\rho_{\Gamma_0,P}(V) \leq c - m$ for margin $m > 0$, then*

$$n \geq \frac{\Gamma_0^2 R^2}{2m^2} \left(\sqrt{\log \frac{2}{\alpha}} + \sqrt{\log \frac{2J}{\beta}} \right)^2 \tag{12}$$

is sufficient for that component to pass with probability at least $1 - \beta/J$. Consequently, if all J components of a fixed candidate have positive margins and satisfy Eq. (12), the candidate passes every component with probability at least $1-\beta$. For an envelope, any $\Gamma_0 > 1$ satisfying these margins yields a nontrivial certified coordinate with the same probability. Exact discrete intervals can improve this sufficient bound but do not weaken it.

Proof. Besides the coverage penalty built into the reported upper curve, a second DKW event at error β/J gives $\widehat{\rho}_{\Gamma_0,n}(V) \leq \rho_{\Gamma_0,P}(V) + \Gamma_0 R \sqrt{\log(2J/\beta)/(2n)}$. Thus the reported upper bound is at most c whenever the sum of this sampling term and the built-in term $\Gamma_0 R \sqrt{\log(2/\alpha)/(2n)}$ is at most m. Solving gives Eq. (12); a union bound over the J power events proves the joint statement. $\square$

Corollary 5 (Minimax prospective evidence allocation). *Fix J components with range lengths R_j, declared budgets Γ_j, positive margins m_j, and component undercoverage probabilities α_j. For joint power $1-\beta$, define the sufficient evidence scores*

$$A_j = \frac{\Gamma_j^2 R_j^2}{2m_j^2}\left(\sqrt{\log\frac{2}{\alpha_j}} + \sqrt{\log\frac{2J}{\beta}}\right)^2. \tag{13}$$

Prospective counts $n_j \geq A_j$ for every component are sufficient for all components to pass with probability at least $1-\beta$. More generally, among positive real allocations with $\sum_j n_j = N$, the allocation minimizing the largest normalized evidence deficit $\max_j A_j/n_j$ is

$$n_j^\star = N\frac{A_j}{\sum_{\ell=1}^J A_\ell}, \qquad \min_{\boldsymbol{n}}\max_j \frac{A_j}{n_j} = \frac{\sum_j A_j}{N}. \tag{14}$$

Thus, when confidence factors and ranges are shared, the minimax sufficient-evidence score is proportional to Γ_j^2/m_j^2. Replacing unknown margins by estimates from an independent design fold, imposing prespecified evidence floors, and rounding to integer counts can change power but not certification validity: conditional on the design fold, the allocation remains fixed before certification. If one sampled evidence cell informs several components, replace its score by the maximum A_j among components sharing that cell and allocate over cells. Meeting the cellwise maxima meets every componentwise sufficient condition, and the same minimax argument applies to these aggregated scores.

Proof. The joint sufficient condition follows by applying Theorem 5 to each component and union bounding the J power events. For any allocation, if $r = \max_j A_j/n_j$, then $A_j \leq rn_j$ for every j, so $\sum_j A_j \leq rN$. Hence $r \geq \sum_j A_j/N$. The displayed allocation makes every ratio A_j/n_j equal to this lower bound. Conditional validity under a design-dependent allocation follows from the probability-space contract in Theorem 1; averaging the conditional guarantee gives the same unconditional guarantee. □

Corollary 5 treats each contract as depending on one evidence cell. Balanced leakage is additive across two source-class cells, and one source-class sample can simultaneously evaluate several attackers. The next result handles both directions of sharing.

Theorem 6 (Additive multi-cell evidence allocation). *Let $c = 1,\ldots,C$ index independently sampled evidence cells and $j = 1,\ldots,J$ index contracts. Contract j has nonnegative weights w_{jc}, population risk*

$$R_j = \sum_{c=1}^C w_{jc}\rho_{\Gamma_{jc},P_{jc}}(V_{jc}), \qquad R_j \leq q_j - m_j, \tag{15}$$

where $m_j > 0$, V_{jc} has range length R_{jc}, and terms with $w_{jc} = 0$ are omitted. Samples inside a cell may be reused across contracts and need not produce independent audit variables. Let D be the number of distinct nonzero cell-risk curves, assign each distinct curve one DKW undercoverage probability α_{jc} (shared wherever that curve is reused), require the sum of those D distinct-curve probabilities to be at most the global certification error budget δ, and define

$$b_{jc} = \frac{w_{jc}\Gamma_{jc}R_{jc}}{\sqrt{2}}\left(\sqrt{\log\frac{2}{\alpha_{jc}}} + \sqrt{\log\frac{2D}{\beta}}\right). \tag{16}$$

If prospective cell counts satisfy

$$\sum_{c=1}^C \frac{b_{jc}}{\sqrt{n_c}} \leq m_j \quad \textit{for every } j, \tag{17}$$

then every contract passes its additive DKW upper bound with probability at least $1-\beta$.

For any positive design-fold margins $\widetilde{m}_j$, put $a_{jc} = b_{jc}/\widetilde{m}_j$. Given budget N and positive floors ℓ_c with $N \geq \sum_c \ell_c$, the prospective surrogate allocation is the convex program

$$\begin{aligned} \underset{\boldsymbol{n},t}{\text{minimize}} \quad & t \\ \textit{subject to} \quad & \sum_c a_{jc} n_c^{-1/2} \leq t \qquad \forall j, \\ & \sum_c n_c = N, \qquad n_c \geq \ell_c \quad \forall c. \end{aligned} \tag{18}$$

It exactly minimizes the worst normalized additive DKW slack for the fixed design-fold coefficients. With one contract and inactive floors, its solution is

$$n_c^\star = N\frac{a_{1c}^{2/3}}{\sum_d a_{1d}^{2/3}}, \qquad \min_{\boldsymbol{n}}\sum_c \frac{a_{1c}}{\sqrt{n_c}} = \frac{(\sum_c a_{1c}^{2/3})^{3/2}}{\sqrt{N}}. \tag{19}$$

If every contract depends on exactly one cell, every optimized cell appears in at least one contract, and the floors are inactive, define $\bar{a}_c = \max_{j:w_{jc}>0} a_{jc}$. Then $n_c^\star = N\bar{a}_c^2/\sum_d \bar{a}_d^2$, recovering the shared-cell square-score allocation in Corollary 5. Design-fold margin substitution, floors, and deterministic integer rounding can change power and exact optimality after rounding, but not certification validity.

Proof. For every distinct nonzero cell-risk curve represented by a pair (j,c), the DKW coverage penalty contributes

$$w_{jc}\Gamma_{jc}R_{jc}\sqrt{\frac{\log(2/\alpha_{jc})}{2n_c}}$$

to the reported additive upper bound. A second DKW event at error β/D upper-bounds the corresponding empirical robust risk above its population value by

$$\Gamma_{jc}R_{jc}\sqrt{\frac{\log(2D/\beta)}{2n_c}}.$$

A union bound over the D power events is valid despite shared samples or dependent audit variables. On their intersection, the reported bound for contract j is at most $R_j + \sum_c b_{jc}/\sqrt{n_c}$, which is no greater than q_j under Eq. (17). The distinct-curve α budget separately makes all reported additive upper bounds valid with probability at least $1-\delta$;

reuse of one curve across contracts spends its undercoverage probability only once.

For fixed positive coefficients, $n \mapsto n^{-1/2}$ is convex on $(0, \infty)$. Each contract constraint in Eq. (18) is therefore convex, and the budget and floor constraints are affine. The epigraph program exactly represents the maximum normalized slack and attains a minimizer on its compact feasible set. With one contract and inactive floors, the Lagrange stationarity condition is $-a_{1c}/(2n_c^{3/2}) + \nu = 0$, so $n_c \propto a_{1c}^{2/3}$; normalizing by N and substituting gives Eq. (19). If contract j uses only cell $c(j)$ and every cell is represented, the objective is $\max_c \bar{a}_c/\sqrt{n_c}$. Squaring it gives the normalized deficit problem with scores $\bar{a}_c^2$, so Corollary 5 yields the last formula when the floors are inactive. Conditioning on the independent design fold fixes all allocation choices before certification; validity follows from the same conditional probability argument used above. □

Theorem 7 (Prospective multi-component lower bound)**.** *For $j = 1, \dots, J$, choose $0 < p_{0j} < p_{1j} < 1$ such that $\Gamma_j p_{1j} \le 1$, $\Gamma_j p_{0j} \le c_j$, and $\Gamma_j p_{1j} > c_j$. A prospective protocol observes n_j independent Bernoulli samples from each component. Let P_0 denote the world in which all components have parameters p_{0j}, and let P_j differ only by replacing p_{0j} with p_{1j}. If a possibly randomized protocol accepts with probability at least $1 - \beta$ under P_0 and falsely accepts with probability at most α under every P_j, where $\alpha + \beta < 1/2$, then*

$$n_j \ge \frac{\log\{1/[2(\alpha + \beta)]\}}{d_{\mathrm{KL}}(\mathrm{Bern}(p_{0j}) \| \mathrm{Bern}(p_{1j}))} \quad \textit{for every } j. \tag{20}$$

Consequently, the total evidence $N = \sum_j n_j$ is at least the sum of these bounds. Fix $\kappa \in (0, 1/2)$. Taking $p_{0j} = (c_j - m_j)/\Gamma_j$ and $p_{1j} = (c_j + m_j)/\Gamma_j$ in a compact interior regime where both probabilities lie in $[\kappa, 1 - \kappa]$ requires

$$N = \Omega\left(\log \frac{1}{\alpha + \beta} \sum_{j=1}^{J} \frac{\Gamma_j^2}{m_j^2}\right). \tag{21}$$

Thus, locally in such compact interior regimes, the quadratic shift-budget and inverse-margin allocation in Corollary 5 is unavoidable up to confidence and κ-dependent constants for prospective independent-cell designs. This scaling statement is not uniform as probabilities approach zero or one; the exact KL bound above remains the global statement.

Proof. Let A be the event that the protocol accepts. The two testing errors satisfy $P_0(A^c) \le \beta$ and $P_j(A) \le \alpha$. The Bretagnolle–Huber inequality applied to P_0 and P_j gives

$$P_0(A^c) + P_j(A) \ge \tfrac{1}{2} \exp\{-n_j\, d_{\mathrm{KL}}(p_{0j} \| p_{1j})\}, \tag{22}$$

because the two product laws differ in only the n_j observations from component j. Rearranging proves Eq. (20) for every j, and summing gives the total lower bound. On $[\kappa, 1-\kappa]$, Bernoulli KL is bounded above by a κ-dependent constant times $(p_{1j} - p_{0j})^2 = (2m_j/\Gamma_j)^2$, proving the stated local order. Near the boundary, that uniform quadratic comparison need not hold, which is why the theorem retains the exact KL denominator. □

Theorem 8 (Unsupported-cell impossibility)**.** *Fix an edit e, a contract threshold c, and let O_n contain every random quantity available to a certification-data-only protocol. Suppose an audit cell A has zero certification probability, the allowed deployment class contains a law Q_A supported on A, and the nonparametric model class contains two worlds M_0, M_1 such that*

1. *O_n has the same distribution under M_0 and M_1;*
2. *$\mathbb{E}_{Q_A, M_0}[V_e] \le c$; and*
3. *$\mathbb{E}_{Q_A, M_1}[V_e] > c$.*

If a possibly randomized protocol controls false acceptance at level δ uniformly over this model and deployment class, then its probability of accepting e under the safe world M_0 is at most δ.

Proof. Include the protocol's independent random seed in O_n. By condition 1, its acceptance probability p is identical in the two worlds. Under M_1 and Q_A, acceptance of e is a false acceptance by condition 3. Uniform control therefore requires $p \le \delta$. The same bound holds under M_0, although condition 2 says that e is safe there. Equivalently, the two observable experiments have total-variation distance zero, so no test can distinguish them. □

An audit cell may be an input region, an environment, or an environment–source-label pair. Thus the result covers both an absent deployment environment and a conditional leakage contract whose required label cell is unobserved. A single-class slice cannot identify the omitted class's conditional leakage without a structural assumption. In the preregistered Camelyon17 split, official hospital center 2 occurs only in the external split, whereas centers 0, 1, 3, and 4 are observed during certification; moreover, the binary center-0 source label is constant in the center-2 external slice. A uniform certificate over the missing cells must acquire representative support, assert a testable cross-cell model, or abstain. The dataset instantiates the support boundary; it is not evidence for the theorem by itself.

4 Exact Finite-Sample Study

The exact study was locked before execution. Its grid is

$$\begin{aligned} &n \in \{250, 500, 1000, 2000, 5000, 10000\}, \\ &\delta \in \{0.01, 0.05, 0.10\}, \qquad \Gamma \in \{1, 1.01, 1.25\}. \end{aligned}$$

It uses 2,000 independent replicates in each of 54 cells. Twelve candidates trade paired target harm in three environments against four attackers, each measured by two source-class recalls. The fixed-profile IUT allocates $\delta/12$ to each candidate; paired harm and balanced leakage split each component allocation between their two exact one-sided bounds.

The finite multinomial/binomial law supplies the exact acceptance probability, simultaneous prediction bands, and false-acceptance bounds. All 54 cells pass: zero false acceptances occur in 108,000 repetitions, and every observed abstention rate falls inside its simultaneous exact prediction band. The largest absolute prediction error is 0.0143. This study validates the discrete implementation under a known law. It does not establish novelty or replace the real benchmark study.

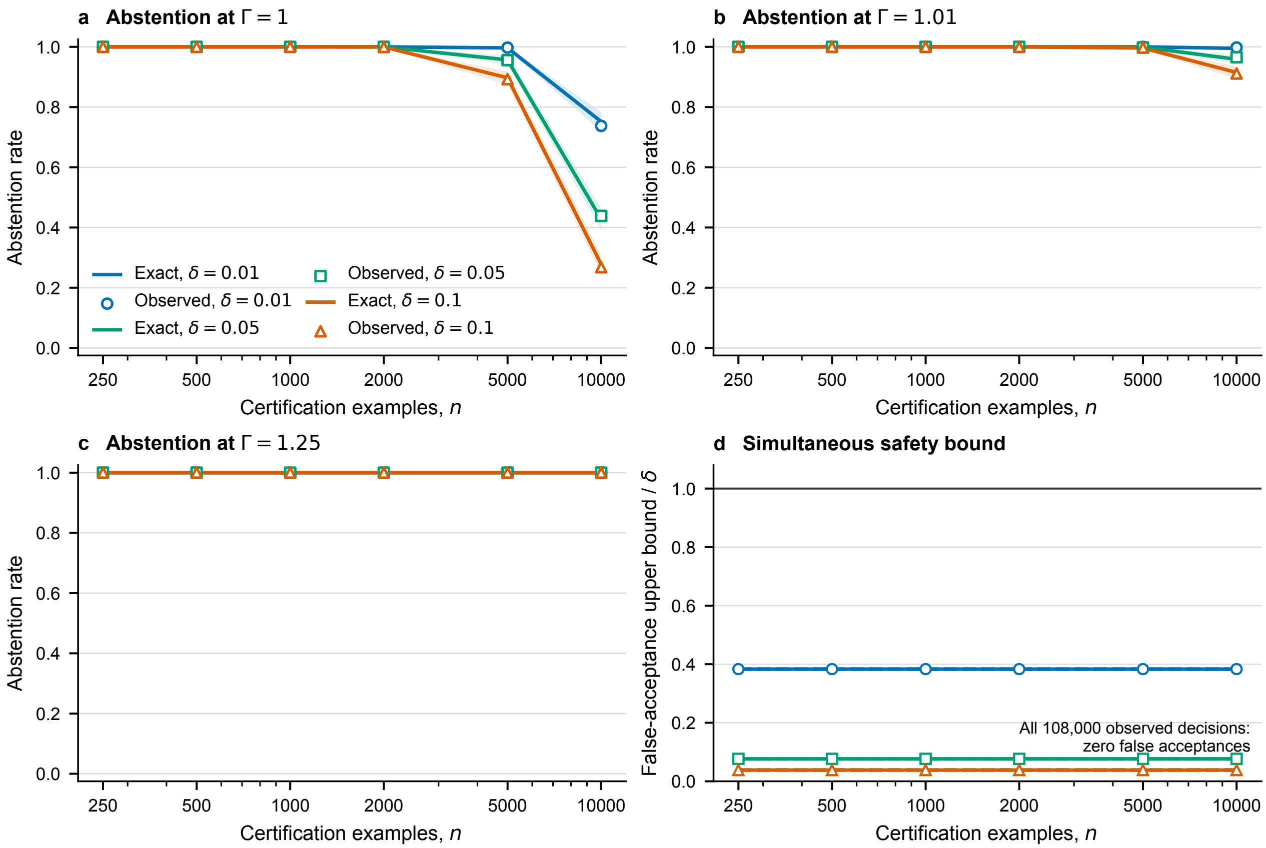

Figure 1: Exact predicted and observed abstention across the 54 locked balanced cells. Bands are simultaneous over the registered grid.

5 Official Real-Study Protocols

The threshold-design study uses untouched seeds 5–12. Its five erasers are INLP, R-LACE, LEACE, TaCo, and MANCE++; its datasets are Waterbirds, Camelyon17-WILDS, CivilComments-WILDS, Bios, and GaitPDB. Candidate construction uses at most 8,000 eraser-training examples and 2,000 construction examples; certification and external evaluation each use at most 8,000 examples. Standardization and randomized PCA are fitted on eraser-training data only. Every edit retrains a balanced logistic target probe and four leakage attackers: linear logistic, RBF-Nystroem logistic, random forest, and a two-layer MLP. Those fresh probes and attackers fit only the eraser-training partition; the separate construction partition is supplied only to pinned edit adapters that require a validation or development input.

The registered grid uses certification fractions $\{0.05, 0.10, 0.25, 0.50, 1.00\}$, paired target-harm thresholds $\tau \in \{0.05, 0.10, 0.20\}$, leakage thresholds $\lambda \in \{0.60, 0.70, 0.80\}$, $\delta = 0.05$, that study's primary budget $\Gamma = 1$, shifted sensitivity $\Gamma = 1.01$, and radius cap $\Gamma_{\max} = 8$. The decision rules are always deploy, point-feasible validation selection, VERA-IUT, the VERA simultaneous envelope, and that study's external oracle used only for evaluation.

Camelyon17 center 2 is absent from certification, while centers 0, 1, 3, and 4 are observed. Its center-0-versus-other binary source label is constant in the center-2 external slice. That earlier balanced protocol therefore assigns center 2 a zero envelope coordinate and abstains for a declared five-center deployment. This is procedural non-certifiability, not by itself a measured external contract violation.

The prospective controlled-shift study addresses a different question under a fully specified finite-reference model. For each of Waterbirds, CivilComments, Bios, and GaitPDB, the official validation audit atoms define P. Independent certification and deployment streams are sampled with replacement from P and from exact registered alternatives Q satisfying $Q_i/P_i \le \Gamma$. The primary cell requests global cap $\Gamma = 1.1$ and uses its exact induced conditional profile, an evidence budget of 4,000, and a targeted allocation with a 15% floor. Secondary cells cross requested global caps $\Gamma \in \{1.1, 1.25, 1.5\}$, budgets $\{1000, 2000, 4000, 8000\}$, and uniform versus targeted allocation. Five official erasers, nine deployment rules, 12 candidate strengths, and 64 fresh seeds yield 1,280 method runs and 3,072 primary candidate evaluations. The deployment outcomes are new draws from a known finite law, not new people, images, or comments; population extrapolation beyond that law remains outside the guarantee.

Dataset	Modality	Target Y	Erased/audited source S	Target environment/support
Waterbirds	vision	bird type	background/place	place; supported
Camelyon17-WILDS	medical vision	tumor status	center 0 vs. centers 1–4	hospital; center 2 unseen
CivilComments-WILDS	text	toxicity	identity mention	identity mention; supported
Bios	text	profession (28-way)	gender label	gender label; supported
GaitPDB	time series	Parkinson status	recording site	recording site; supported

Table 5: Cross-modal protocol and support scope. The four supported rows enter the prospective $\Gamma > 1$ study. Camelyon17 is a separate identification boundary: center 2 is absent during certification, so it cannot enter a supported reweighting law. This is not a clinical claim.

Family	Pinned upstream entry point	Shared-protocol adaptation	Candidate strengths
INLP	`src.debias.get_debiasing_projection`	Official nullspace directions; the removed scikit-learn loss alias `log` is replaced by its `log_loss` name.	ranks 1, 2, 4, 8
R-LACE	`rlace.solve_adv_game`	Official adversarial projection with the same loss-name compatibility change; train/construction splits replace the upstream train/dev inputs.	ranks 1, 4
LEACE	`concept_erasure.LeaceFitter.fit`	Closed-form `method=leace` map fitted to one-hot source labels on eraser-training representations.	one closed-form map
TaCo	`sobol_importance_from_sample`; `crop_concepts`; `build_gender_neutral_features`	PCA and target/source heads are fitted only on eraser-train/construction data; official Sobol scoring and concept cropping build each edit.	remove 1, 2, 3, or 5 of 20 components
MANCE++	`mance.MANCE(...).fit_erase`	Three fixed steps with `stop_at_floor=False`. Dummy deployment labels satisfy the upstream logging interface; pinned source review confirms that labels enter evaluation history, not the edit update.	one $\epsilon = 0.05$ map

Table 6: Official-entry-point parity under the common frozen-feature protocol. The adapters do not claim to reproduce every upstream paper's full training pipeline. Receipts pin the remote, commit, entry point, configuration, split hashes, and edited-array hash, and the audit rejects dirty upstream trees or proxy-marked candidates.

Primary contract status and the exact shifted-law opportunity denominator are evaluated by the exact Q-weighted mean over all finite-reference atoms. The independent 50,000-draw deployment stream is retained as a Monte Carlo replay and sampling-diagnostic endpoint; it cannot change a primary violation label or candidate selection.

Allocation operates over target-environment and source-class evidence cells. A candidate is selected on the independent design fold by the registered largest-minimum-normalized-margin rule. A source-class draw evaluates every registered attacker, so its score uses the worst attacker leakage margin for that selected candidate; it does not pretend that attacker-specific samples are independent. Thus the allocation targets the stated sufficient-evidence objective for one design-fold candidate, while the subsequent certificate remains valid for every registered candidate. The registered primary uses the shared-cell square-score construction in Corollary 5; because balanced leakage adds two source-class cells, the prospectively frozen extension solves the full additive convex program in Theorem 6. It uses disjoint certification streams and cannot alter any primary decision.

The dataset-specific target and leakage thresholds were selected before these fresh seeds by a deterministic Cartesian-grid search on the completed seeds 5–12 block, which was used only as design data for the later disjoint blocks. The design first sought zero certificate-selected external violations and at least 20% point-selection violations, then used fixed utility and threshold tie-breaks. The selected thresholds were frozen before the disjoint seeds 13–44 replication and inherited unchanged by seeds 45–108. They are benchmark operating points for comparing matched decision rules, not clinical, legal, or stakeholder-derived definitions of acceptable harm or leakage.

6 Statistical Integrity

For the earlier seeds 5–12 threshold-design study, threshold pairs and nested fractions share the same eight seed-level fits and examples. Its configuration-level Clopper–Pearson and McNemar calculations are retained as preregistered diagnostics, not treated as independent trial inference. That study averages each rule's nine outcomes within seed and applies its locked one-sided sign-flip test over eight seed blocks, with Holm correction over four externally estimable datasets. All adjusted values are reported regardless of outcome. Seeds 0–4 contribute to none of those estimates or tests.

For the prospective controlled-shift study, the seed is the independent cluster. The primary safety endpoint uses a rotating held-out dataset sentinel and the registered one-sided Clopper–Pearson gate for false acceptance. The four rotating dataset laws need not have a common event probability. This does not invalidate the primary decision: at 64 sentinels the gate passes only when there are zero events, and $\Pr(S=0)=\prod_i(1-p_i)\le(1-\bar{p})^{64}$ by AM–GM. Thus its zero-event upper limit is conservative for the average sentinel risk $\bar{p}$ under independence. If any event occurs, the gate fails; nonzero-event binomial intervals are reported only as descriptive diagnostics, alongside a heterogeneity-robust supplementary bound. The separately frozen safety sensitivity also reports a seed-familywise any-dataset-violation bound and per-dataset ordinary one-sided 95% Clopper–Pearson bounds. A simultaneous four-dataset confidence set uses Bonferroni component level $0.05/4$; separate exact lower-tail tests against 5% are Holm adjusted within rule and are not called confidence bounds. The vector rule additionally reports the symmetric within-seed dataset co-occurrence matrix and counts of seeds with zero through four violating datasets. None treats the four decisions within a seed as independent, and none can replace the registered sentinel gate. The primary usefulness endpoint bootstraps whole seed clusters and requires the lower confidence bound for vector-certificate retention to be at least 20%. It uses 20,000 percentile-bootstrap replicates and random seed 2027071601. The registered paired comparison aggregates all four datasets within each seed and uses an exact sign test; per-dataset secondary tests use Holm correction. The vector/common point ratio remains the registered gate. A frozen supplementary interval retains zero-common bootstrap samples and also reports the division-free $(V-2C)/O$ contrast, where V and C are vector and common safe-retention counts and O is the opportunity count; it cannot change the gate. Its 20,000 resamples are partitioned exactly once among $(O>0, C>0)$, $(O>0, C=0, V>0)$, $(O>0, C=0, V=0)$, and $(O=0, V=C=0)$; a positive-infinite extended-ratio endpoint is serialized explicitly rather than clipped or dropped. All confidence intervals, adjusted values, limiting coordinates, and effect sizes are reported whether favorable or not.

Three-rule contract-severity stress. Using the same candidates, arrays, exact shifted laws, confidence bounds, and utility tie-break, a predefined sensitivity recomputes always deploy, validation point selection, and the vector envelope at $\kappa\in\{0.75, 1, 1.25\}$. Dataset thresholds are $\tau_d(\kappa)=\kappa\tau_d$ and $\lambda_d(\kappa)=0.5+\kappa(\lambda_d-0.5)$, so $\kappa=1$ is exactly the registered contract. Every rule–dataset–severity row is reported with 64 all-decision trials, deployment and deployed-only violation rates, exact shifted-law safe opportunities, retention, and whole-seed intervals. A supporting main-track headline is allowed only if both naive rules separately reach 20% violations on a registered-threshold dataset, VERA remains at or below 5% in every cell, and the registered safety and usefulness gates pass. Failure is a required negative result and cannot be rescued by another severity; this sensitivity does not alter the primary decision.

External labels are excluded from edit construction and candidate selection. The runner computes external arrays only after each candidate is fixed; those arrays enter post-selection evaluation, not construction. This is a procedural separation rather than a claim that labels were physically inaccessible during execution.

7 Controlled-Shift Results

The sealed controlled-study result layer is intentionally absent from this outcome-blind source build.

8 Claim Boundary

VERA does not prove perfect erasure, security against every measurable attacker, fairness for unmeasured concepts, clinical safety, or validity on new support. The density-ratio budgets are declared sensitivity parameters, not facts estimated from an external test set. Receipt consistency and code replay do not by themselves establish novelty or proof correctness.

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